LifeLink Platform

End-to-End Cross-Portal Verification & Operational Testing Guide

Document Version: 2.0.0 | **Release Status:** Production Ready | **Portals:** Donor, Hospital, Blood Bank

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1. System Environments & Test Accounts

LifeLink is an integrated blood supply chain and transfusion management system. It coordinates donor eligibility verification, phlebotomy collections, component fractionation, infectious disease laboratory testing, hospital clinical requisitions, and cold-chain courier dispatch across three specialized web portals.

Application Role	Local URL / Access	Backend Target API
Donor Portal	http://localhost:5173	http://localhost:5001/api/donors
Hospital Portal	http://localhost:5173	http://localhost:5001/api/hospitals
Blood Bank Portal	http://localhost:5173	http://localhost:5001/api/blood-bank

Pre-Configured Demonstration Accounts

Portal Role	Email Address	Password	Organization / Purpose
Blood Bank Staff	admin@redcrossblood.org	password123	Metropolitan Red Cross Center
Hospital Staff	stjudes@metrohealth.org	password123	St. Jude Children's Hospital
Verified Donor	sarah.j@example.com	password123	Sarah Jenkins (A+ Donor)
New Unverified	alex.rivera@example.com	password123	Newly Registered Test Donor

2. Cross-Portal Verification Scenarios

Scenario 1: Donor Registration, Verification Barrier & Pass Unlock

Target Flow: Enforce medical safety verification before enabling active donor pass or donation eligibility.

- **Step 1 (Register):** Navigate to /register, select *Donor* role, register with name Alex Rivera, Blood Group O+, weight 68 kg.
- **Step 2 (Verify Ineligible State):** Log into Donor Dashboard (/donor/dashboard). Observe amber warning banner: *'Verification Required: Please upload your blood group report or government ID'*. Digital pass barcode is locked.
- **Step 3 (Document Upload):** Go to /donor/profile. Enter Document: *Lab Blood Report*, ID: LAB-9942, URL: https://cdn.lifelink.io/reports/rivera.pdf. Click Submit. Badge transitions to *'Pending Verification'*.
- **Step 4 (Staff Approval):** In another tab, log into Blood Bank Portal as admin@redcrossblood.org. In donor lookup, find Rivera. Observe badge *'Pending Verification'*. Click **[Verify Donor]**.
- **Step 5 (Confirmation):** Refresh Donor Portal. Amber alert is replaced with green badge: *'Verified Donor — You are eligible to donate blood!'* Digital pass is now unlocked with a verified QR code.

Scenario 2: Phlebotomy Collection & Clinical Cool-Off Window

Target Flow: Record in-house donation, block duplicate same-day entries, and activate mandatory cool-off countdown.

- **Step 1 (Record Collection):** In Blood Bank Portal, go to Collections Hub (/blood-bank/collections) > *'In-House Phlebotomy'* tab. Click **[Record New Collection]**. Select donor Alex Rivera, Component Whole Blood, Volume 450 mL, Phlebotomist Nurse Dean. Click Save.
- **Step 2 (Duplicate Prevention Test):** Click Record New Collection again immediately for the same donor and date. The server strictly rejects the request with HTTP 409 or 422 deferral error.
- **Step 3 (Donor Cool-off Activation):** Return to Donor Portal. Dashboard now indicates *'Ineligible — Cool-off window active'* with an exact 56-day countdown until next permissible donation.

Scenario 3: Phlebotomy Component Fractionation & Quarantine Induction

Target Flow: Separate whole blood into clinical components and hold in quarantine until infectious disease tests pass.

- **Step 1 (Fractionate):** On Collections table, locate the new collection and click **[Fractionate]**. Specify Component 1: Red Blood Cells (RBC) 280 mL in Chiller Shelf 1, and Component 2: Plasma 150 mL in Deep Freeze 2.
- **Step 2 (Quarantine Inspection):** Navigate to Laboratory Screening (/blood-bank/testing). Both units appear in *Pending Testing* quarantine queue.
- **Step 3 (Safety Verification):** Confirm unreleased components remain strictly locked from allocation queries.

Scenario 4: 5-Marker Lab Screening & Auditable Clinical Clearance

Target Flow: Separately enter 5 mandatory disease markers (HIV, HBV, HCV, Syphilis, Malaria) and authorize clinical clearance.

- **Step 1 (Quarantine Inspection):** Navigate to Laboratory Screening (/blood-bank/testing) or click a quarantined unit badge from Hospital Requests. The full quarantine queue is loaded dynamically.
- **Step 2 (Individual Marker Recording):** Select the unit. For each of the 5 mandatory markers (HIV 1/2, Hepatitis B, Hepatitis C, Syphilis, Malaria), record actual assay results (*Negative*, *Pending*, or *Reactive*).
- **Step 3 (Safety Gate Enforcement):** Missing or pending markers strictly block clinical clearance with HTTP 422. Reactive markers trigger immediate biohazard quarantine.
- **Step 4 (Authorization & Stock Release):** When all 5 markers are confirmed Negative, click **[Authorize & Release Unit to Available Stock]**. Status updates to *'Passed'*, and unit immediately enters eligible FEFO inventory.

Scenario 5: Hospital Requisition & Diagnostic FEFO Allocation

Target Flow: Match compatible units using First-Expired-First-Out, with quarantine diagnostic alerts.

- **Step 1 (Hospital Request):** In Hospital Portal (stjudes@metrohealth.org), go to /hospital/requisitions. Click **[Create Blood Requisition]**: Patient Baby Thomas (Pediatric), Blood Group O+, Component RBC, Units: 1, Urgency: Urgent.
- **Step 2 (Diagnostic Allocation Hub):** In Blood Bank Portal, open Hospital Requests Hub (/blood-bank/requests). Open requisition drawer. The system runs FEFO matching.

- **Step 3 (Quarantine Alert & Direct Jump):** If compatible units are still untested, an amber card alerts: *'Compatible units held in quarantine awaiting lab screening'* with clickable badges taking you directly to screen that unit.
- **Step 4 (Barcode Assignment & Reservation):** Enter or select the cleared unit and click **[Allocate & Reserve Units]**. Status updates to *'Reserved'*, and the hospital receives a real-time notification.

Scenario 6: Cold Chain Dispatch & Hospital Receipt Confirmation

Target Flow: Validated courier handoff with temperature logger and hospital receipt verification.

- **Step 1 (Dispatch Preparation):** On Blood Bank Requisitions Hub, click **[Prepare Dispatch]**. Enter Courier: Dedicated Hospital Transport, Vehicle: VAN-METRO-09, Seal: SEAL-9081, Temp Range: 2.0°C - 6.0°C.
- **Step 2 (Courier Handoff):** Click **[Confirm Courier Handoff]**. Status transitions to *'Dispatched'*.
- **Step 3 (Hospital Receipt):** In Hospital Portal, find the dispatched order and click **[Confirm Delivery & Receive Units]**. Enter Received Temp: 3.6°C and Seal Intact: Yes. Order marks *'Fulfilled'*; blood units mark *'Issued'*.

Scenario 7: Organizational Donation Camp Request & Drive Scheduling

Target Flow: Corporate/community drive request submission, review, approval, and live camp scheduling.

- **Step 1 (Submit Camp Request):** In Donor Portal, navigate to /donor/camps > **[Organize a Blood Drive]**. Organization: Metro Tech Campus, Expected Donors: 75, Target Facility: Metropolitan Red Cross.
- **Step 2 (Blood Bank Review):** In Blood Bank Portal, go to Collections Hub > *'Camp Requests'* tab. View the pending request from Metro Tech Campus.
- **Step 3 (Accept & Schedule):** Click **[Accept & Schedule Drive]**. Set Goal Units: 60, Hours: 09:00 AM - 04:00 PM, click Confirm. Request updates to *'Approved'*, a scheduled camp appears on public donor map, and the organizer is notified.

Scenario 8: Authentic Analytics & Report Validation

Target Flow: Validate dynamic report calculations and confirm complete removal of hardcoded mock numbers.

- **Step 1 (Reports View):** Navigate to Blood Bank Operational Reports (/blood-bank/reports).
- **Step 2 (Verification):** Confirm that Collections This Month and Hospital Requests reflect true counts (not fallback constants 124 or 88). Confirm fulfillment rate is dynamically computed as (fulfilled / total) * 100% (not hardcoded 94.2%).

3. Automated CLI Verification Suites

LifeLink includes comprehensive automated integration test suites executable from the terminal. Run these commands from the repository root to verify all core workflows:

Command	Tests	Coverage Domain
`node ./frontend/src/tests/labScreeningAndAllocation.test.mjs`	10 / 10	Quarantine healing, individual viral markers, 422 safety gates, allocation
`node ./frontend/src/tests/workflowAuditAndRepairs.test.mjs`	5 / 5	Audit fixes, zero metrics, FEFO diagnostics, camp approval, alerts
`node ./frontend/src/tests/donorEligibilityVerification.test.mjs`	11 / 11	Eligibility barriers, spoofing blocks, 422 deferrals, documents

`node ./frontend/src/tests/cleanEndToEndWorkflow.test.mjs`	8 / 8	Full 3-portal end-to-end lifecycle starting from clean state
`node ./frontend/src/tests/hospitalSuite.test.mjs`	50 / 50	Hospital requisitions, priority matching, emergency logistics
`node ./frontend/src/tests/fulfillmentSuite.test.mjs`	15 / 15	FEFO allocations, cold storage, courier handoff, hospital receipt

4. Operational FAQs & Recovery Procedures

Q: Why does a donor show 'Ineligible' immediately after a blood collection?
A: Standard clinical guidelines mandate a cool-off window (56 days for Whole Blood, 90 days for RBC). The dashboard countdown reflects remaining days until the donor can donate again.

Q: How do I restore pristine demo fixtures for presentations?
A: Send a POST request to http://localhost:5001/api/test/seed-fixtures via curl or the developer options panel in the application settings.

Q: What happens if browser geolocation permissions are denied?
A: The application implements a three-tier location hierarchy: (1) Saved Profile City, (2) Interactive Manual Search Bar, (3) GPS fallback. Denied permissions will never block user workflows.